

Predicting property prices

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Problem and Applications

Predicting real estate prices is a classic machine learning task. It involves estimating prices based on historical data.

Applications include:

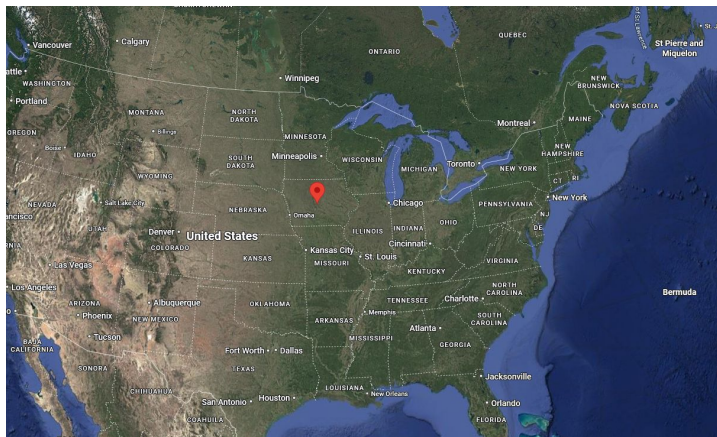
1. Quick valuations for banks and other institutions
2. Identifying investment opportunities
3. Dynamic pricing for developers
4. Analysis of factors affecting price

Dataset

I used the **Ames Housing Dataset** from Kaggle.

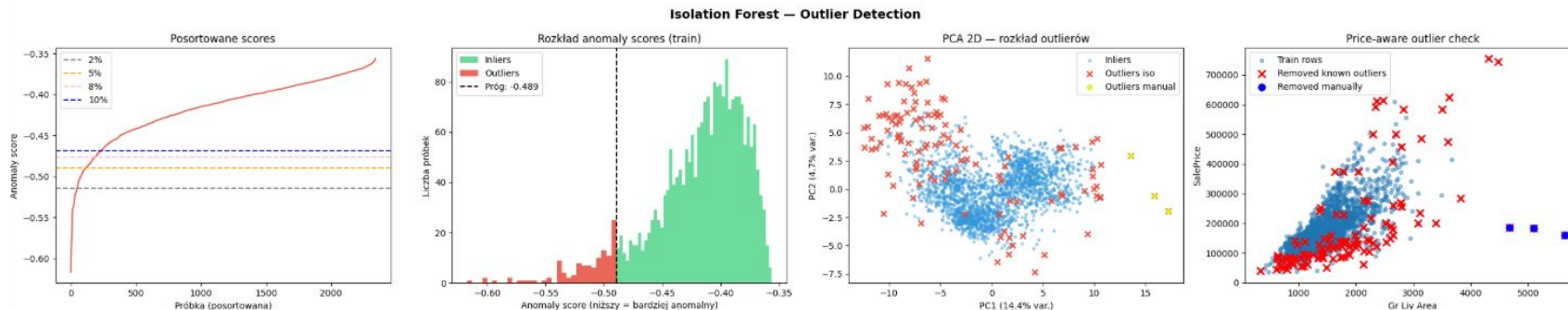
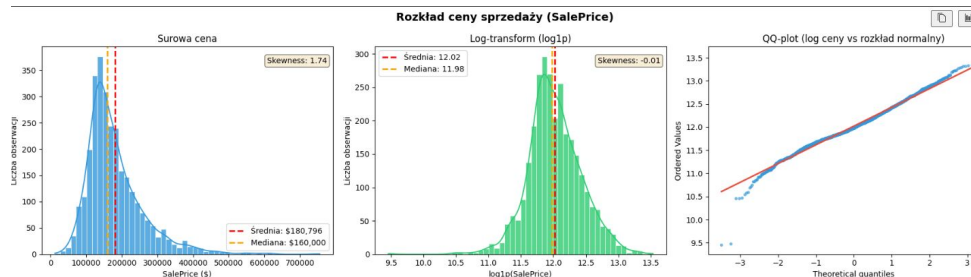
It contains 2,930 property data points, each with 79 features (floor area, number of rooms, foundation type, location, etc.).

The data comes from the city of Ames in Iowa, USA.

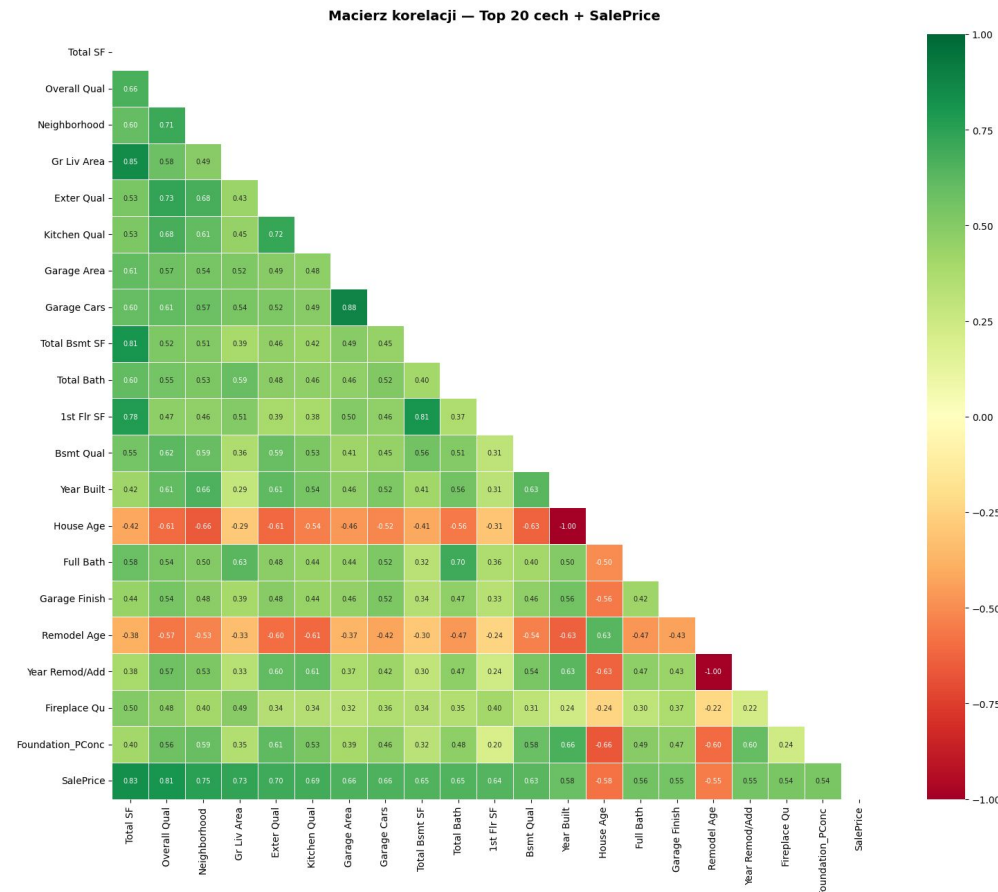


Data Preparation

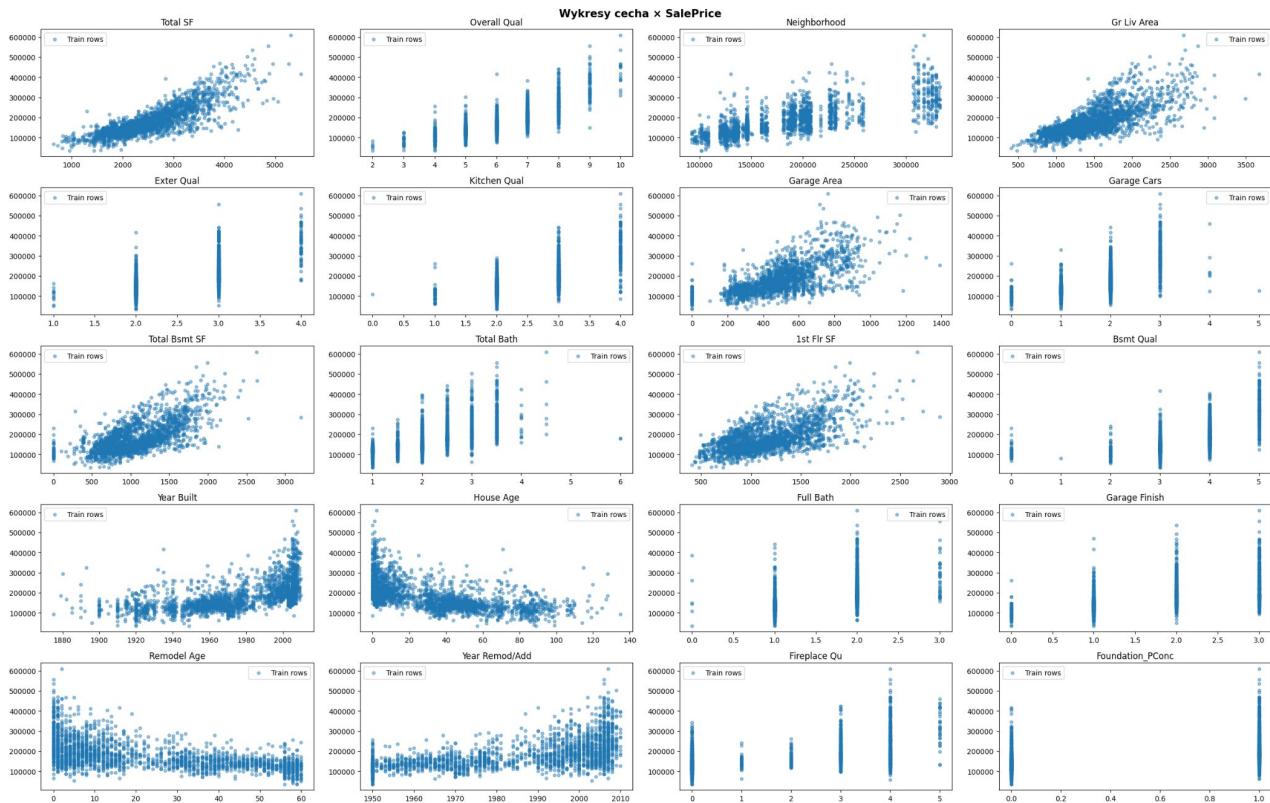
1. Handling Missing Values
2. Feature Engineering (adding new)
3. Price Log Transform
4. Splitting into Training and Test Sets
5. Encoding Text Values
6. Removing Outliers
7. Scaling



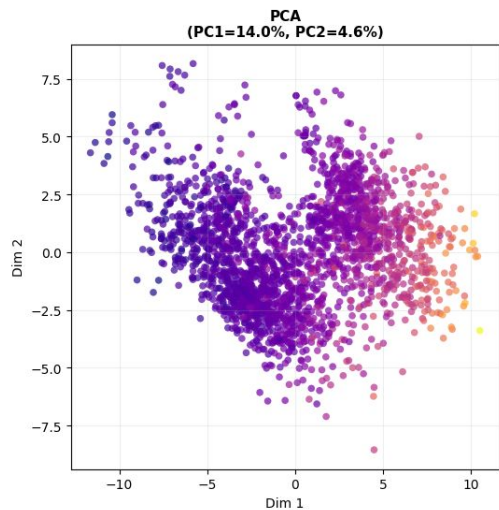
Data Visualization



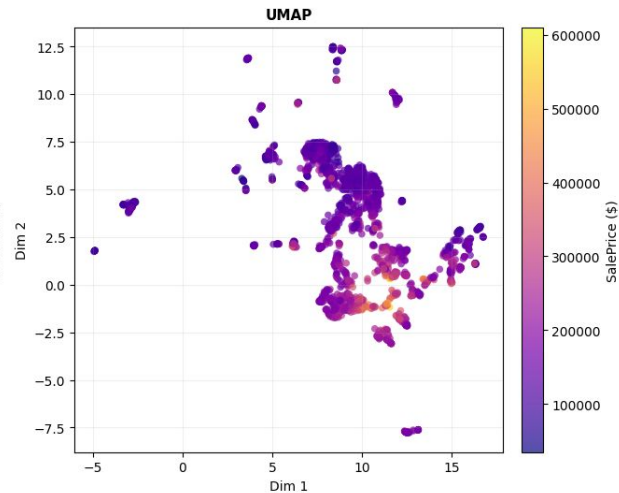
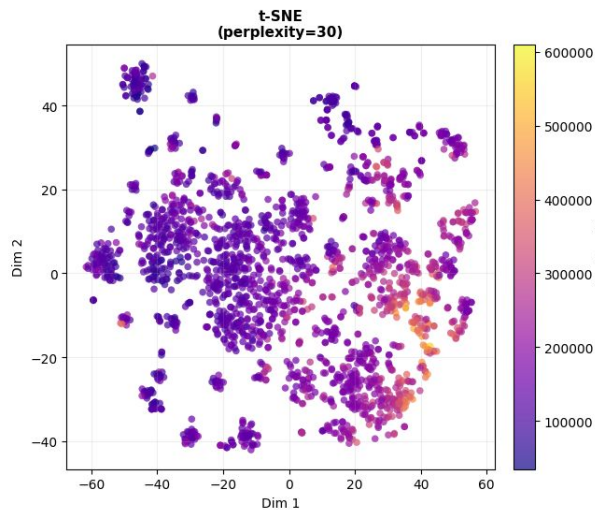
Data Visualization



Data Visualization



Redukcja wymiarów do 2D — kolorowane SalePrice



Models

**Linear
Regression**

Decision Tree

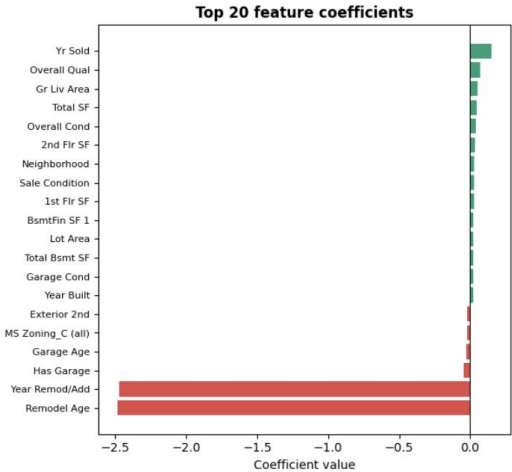
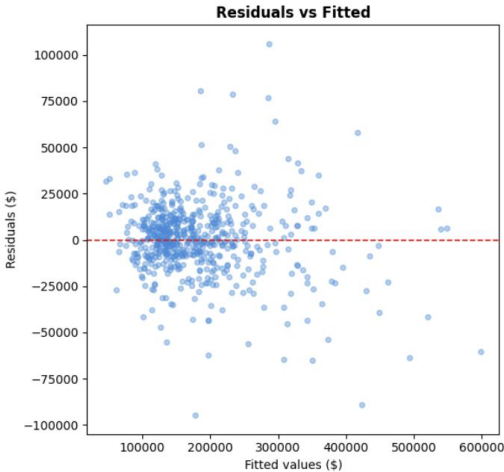
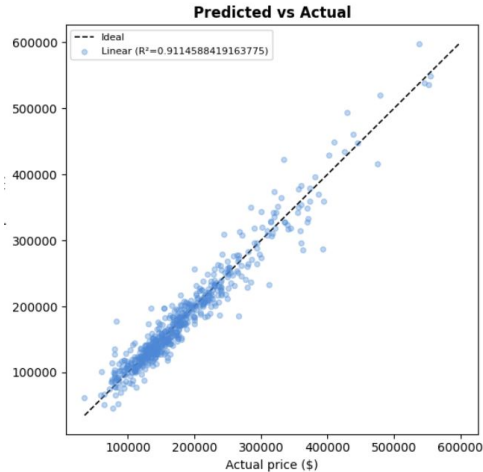
**Polynomial
Regression**

**More advanced models
(Random Forest, HistGradientBoosting, XGBoost)**

Linear Regression

	RMSE \$	MAE \$	R ² (Log)	RMSLE
Linear Regression	19761	13734	0.9115	0.1194

Linear Regression Model - Ames Housing



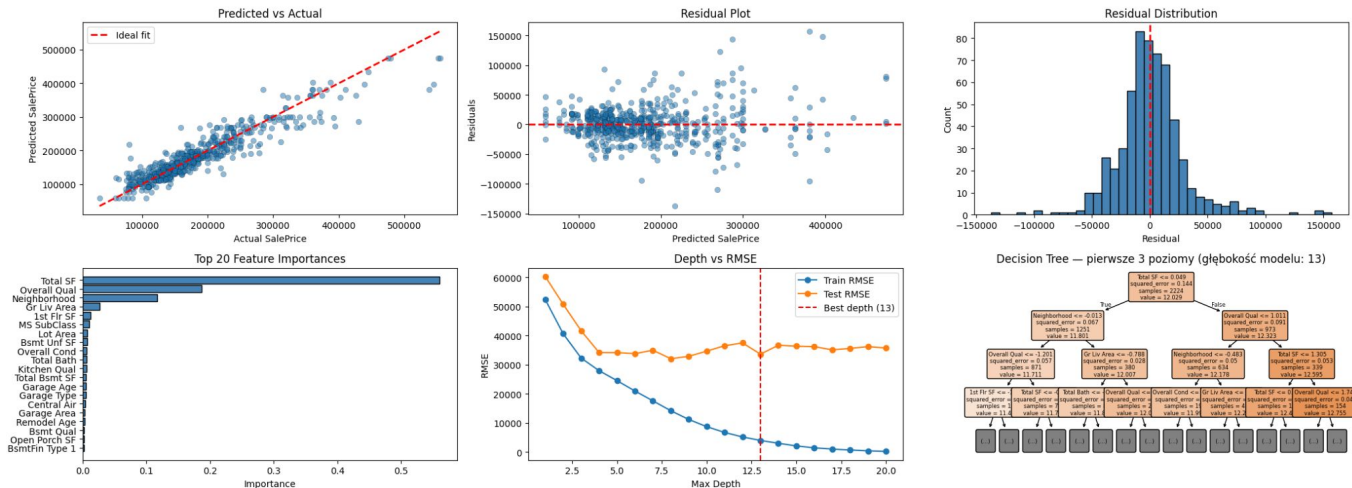
Decision Tree

Tree best params:

```
{'ccp_alpha': 0.0, 'max_depth': 15, 'max_features': None, 'min_samples_leaf': 10, 'min_samples_split': 2}
```

	RMSE \$	MAE \$	R ² (Log)	RMSLE
Decision Tree	29506	20374	0.8327	0.1642

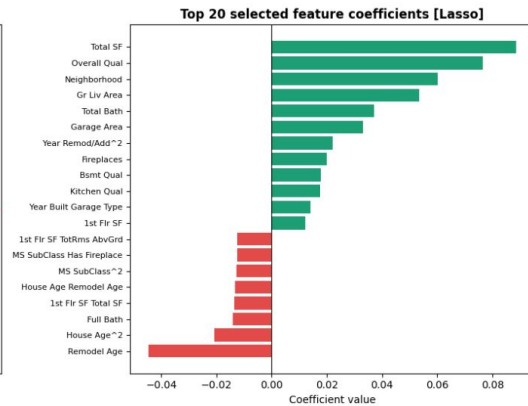
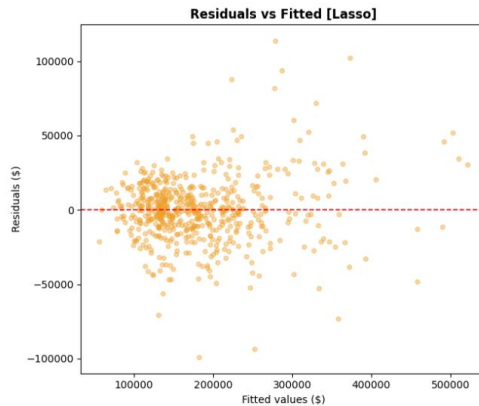
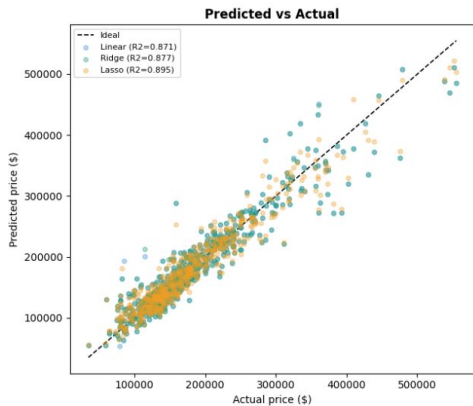
Decision Tree — Analiza modelu



Polynomial Regression

	RMSE \$	MAE \$	R ² (Log)	RMSLE
Normal	25581	17235	0.8711	0.1441
Ridge ($\alpha = 0.01$)	25307	17163	0.8769	0.1408
Lasso ($\alpha = 0.001$)	22024	15536	0.8949	0.1301

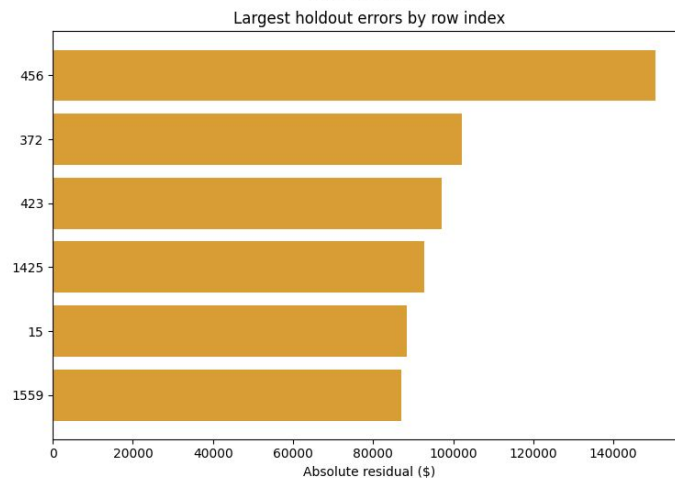
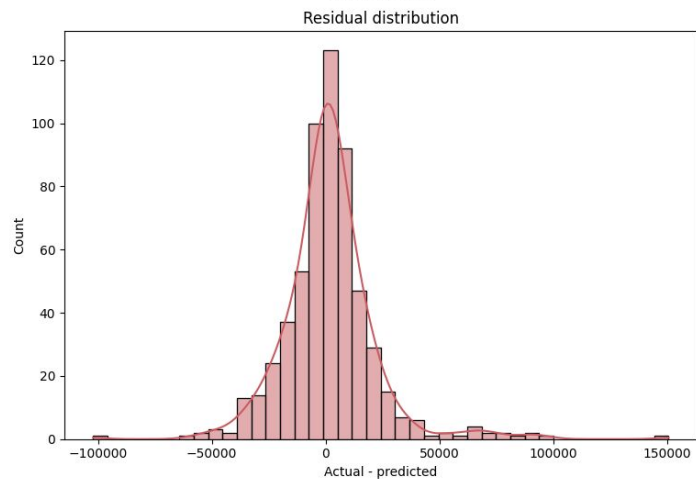
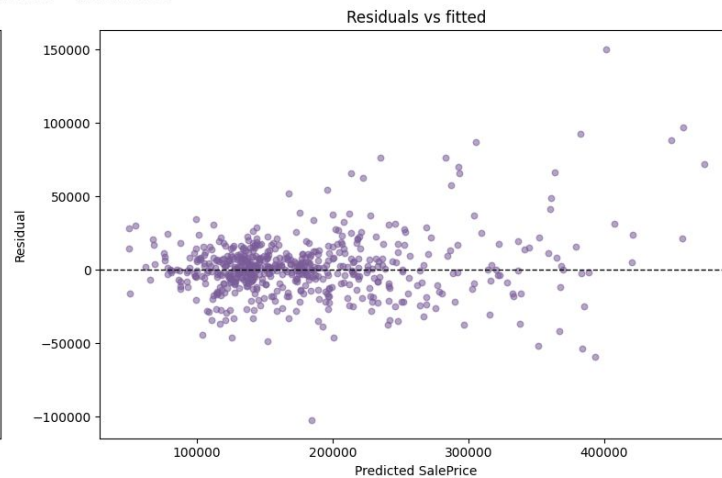
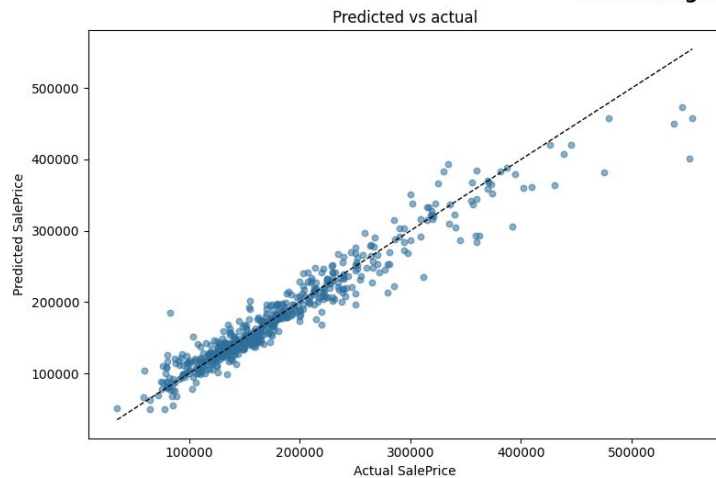
Polynomial Regression Models - Ames Housing



Advanced models

	RMSE \$	MAE \$	R ² (log)	CV R ²	RMSLE	CV RMSLE
XGBoost	20677.56	13414.00	0.920	0.918	0.113	0.109
HistGradientBoosting	21575.39	13671.16	0.918	0.916	0.115	0.110
Polynomial Lasso	22023.98	15535.89	0.895	0.898	0.130	0.121
RandomForest	26666.77	16010.54	0.889	0.895	0.134	0.123
Linear	19761.25	13733.60	0.911	0.857	0.119	0.134
DecisionTree	29289.72	19941.20	0.828	0.818	0.166	0.161

Model diagnostics - XGBoost



SUMMARY